

IGCSE Chemistry CIE

YOUR NOTES



3. Stoichiometry

CONTENTS

3.1 Formulae & Relative Masses

3.1.1 Formulae

3.1.2 Empirical Formulae & Formulae of Ionic Compounds

3.1.3 Writing Equations

3.1.4 Ar & Mr

3.2 The Mole & the Avogadro Constant

3.2.1 The Mole

3.2.2 Linking Moles, Mass & Mr

3.2.3 Reacting Masses

3.2.4 Calculating Concentration

3.2.5 Titration Calculations

3.2.6 Empirical & Molecular Formula

3.2.7 Percentage Yield & Purity

3.1 Formulae & Relative Masses

3.1.1 Formulae

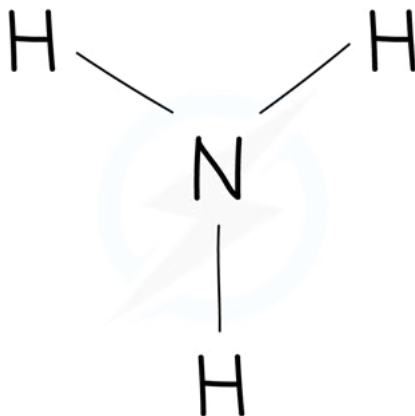
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Molecular Formulae

Element symbols

- Each element is represented by its own unique symbol as seen on the Periodic Table
 - E.g. H is hydrogen
- Where a symbol contains two letters, the first one is always in **capital** letters and the other is **small**
 - E.g. sodium is Na, not NA
- Atoms combine together in **fixed ratios** that will give them full outer shells of electrons
- The chemical formula tells you the ratio of atoms
 - E.g. H_2O is a compound containing 2 hydrogen atoms which combine with 1 oxygen atom
- The chemical formula can be deduced from the relative number of atoms present
 - E.g. If a molecule contains 3 atoms of hydrogen and 1 atom of nitrogen then the formula would be NH_3
- Diagrams or models can also be used to represent the chemical formula



- *The ammonia molecule consists of a central nitrogen atom bonded to three hydrogen atoms*

Chemical formulae

- The **structural** formula tells you the way in which the atoms in a particular molecule are bonded
 - This can be done by either a **diagram** (displayed formula) or **written** (simplified structural formula)

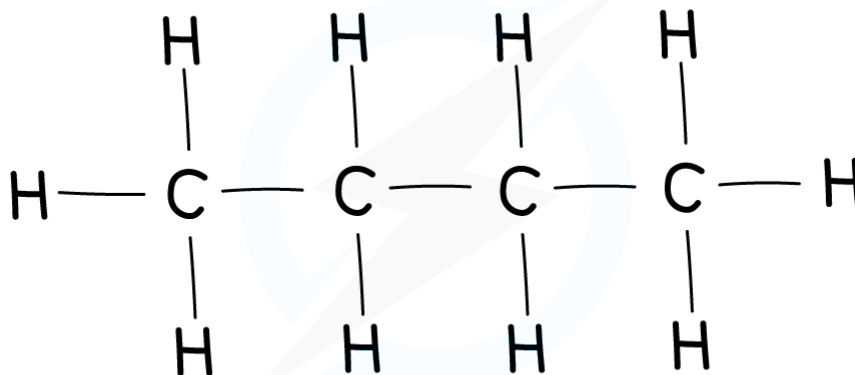
- The **molecular** formula tells you the actual number of atoms of each element in one molecule of the compound or element
 - E.g. H_2 has 2 hydrogen atoms, HCl has 1 hydrogen atom and 1 chlorine atom

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Example: Butane

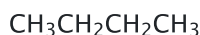
- Structural formula (displayed)



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- Structural formula (simplified)



- Molecular formula



- Empirical formula



Deducing formulae by valency

- The concept of valency is used to deduce the formulae of compounds (either molecular compounds or ionic compounds)
- Valency or combining power tells you how many bonds an atom can make with another atom or how many electrons its atoms lose, gain or share, to form a compound
 - E.g. carbon is in Group IV so a single carbon atom can make 4 single bonds or 2 double bonds
- The following valencies apply to elements in each group:

GROUP	VALENCY
I	1
II	2
III	3
IV	4
V	3
VI	2
VII	1
VIII	0

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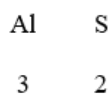


Worked Example

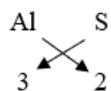
What is the formula of aluminium sulfide?

Answer:

- We can use the valency of each atom to work out a formula



Write out the symbols of each element and write their combining powers underneath



The formula is then calculated by cross multiplying each atom with the number opposite



The formula for aluminium sulfide is Al_2S_3

3.1.2 Empirical Formulae & Formulae of Ionic Compounds

Empirical Formulae**EXTENDED**

The **molecular formula** is the formula that shows the **number** and **type** of each atom in a molecule

- E.g. the molecular formula of ethanoic acid is $\text{C}_2\text{H}_4\text{O}_2$
- The **empirical formula** is the simplest whole number ratio of the atoms of each element present in one molecule or formula unit of the compound
 - E.g. the empirical formula of ethanoic acid is CH_2O
- **Organic molecules**, such as ethanoic acid, often have **different** empirical and molecular formulae
- The formula of an **ionic compound** is always an **empirical formula**

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Deducing Formulae of Ionic Compounds

EXTENDED

- The formulae of these compounds can be calculated if you know the charge on the ions
- The Periodic Table can help work out the charge on many elements:
 - Group I elements form ions with a 1+ charge
 - Group II elements form ions with a 2+ charge
 - Group III elements form ions with a 3+ charge
 - Group V elements form ions with a 3- charge
 - Group VI elements form ions with a 2- charge
 - Group VII elements form ions with a 1- charge
- Below are some other common ions and their charges
- Note that a Roman numeral next to the element tells you the charge on the ion, e.g. copper(II) ions have a charge 2+
- There are several common compound ions included in the table
 - Some chemists call these polyatomic ions

Common Ions & Their Charges Table

Ion	Formula and charge
Iron(II)	Fe^{2+}
Copper(II)	Cu^{2+}
Chromium(III)	Cr^{3+}
Ammonium	NH_4^+
Hydroxide	OH^-
Nitrate	NO_3^-
Sulfate	SO_4^{2-}
Carbonate	CO_3^{2-}
Hydrogen carbonate	HCO_3^-

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- The overall sum of the charges of an ionic compound should be 0

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- You therefore need to work out the ratio of the ions to ensure this is the case
- When you write the formula of a compound ion it is necessary to use brackets around the compound ion where more than one of that ion is needed in the formula
 - For example copper(II) hydroxide is $\text{Cu}(\text{OH})_2$

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? Worked Example

What is the formula of?

- sodium bromide
- aluminium fluoride
- aluminium oxide
- magnesium nitrate
- ammonium sulfate

Answer 1

Symbol	Na	Br
Ion charge	1+	1-
Balance the number of ions	1 sodium ion is needed for each bromide ion	
Ratio of ions	1:1	
Formula	NaBr	

Answer 2

Symbol	Al	F
Ion charge	3+	1-
Balance the number of ions	3 fluoride ions are needed for each aluminium ion	
Ratio of ions	1:3	
Formula	AlF_3	

Answer 3

Symbol	Al	O
Ion charge	3+	2-
Balance the number of ions	2 aluminium ions are needed for 3 oxide ions	
Ratio of ions	2:3	
Formula	Al_2O_3	

Answer 4

Symbol	Mg^{2+}	NO_3^-
Ion charge	2+	1-
Balance the number of ions	2 nitrate ions are needed for each magnesium ion	
Ratio of ions	1:2	

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Formula	$\text{Mg}(\text{NO}_3)_2$
---------	----------------------------

Answer 5

Symbol	NH_4^+	SO_4^{2-}
Ion charge	1+	2-
Balance the number of ions	2 ammonium ions are needed for each sulfate ion	
Ratio of ions	2:1	
Formula	$(\text{NH}_4)_2\text{SO}_4$	

3.1.3 Writing Equations

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Writing Word Equations & Symbol Equations

Word equations

- These show the reactants and products of a chemical reaction using their full chemical names
- The arrow (which is spoken as “goes to” or “produces”) implies the conversion of reactants into products
- Reaction conditions or the name of a catalyst can be written above the arrow
- An example of a word equation for neutralisation is:

sodium hydroxide + hydrochloric acid → sodium chloride + water

- The reactants are sodium hydroxide and hydrochloric acid
- The products are sodium chloride and water

Names of compounds

- For compounds consisting of 2 atoms:
 - If one is a **metal** and the other a **non-metal**, then the name of the metal atom comes first and the ending of the second atom is replaced by adding **-ide**
 - E.g. NaCl which contains sodium and chlorine thus becomes sodium **chloride**
 - If both atoms are **non-metals** and one of those is **hydrogen**, then hydrogen comes first
 - E.g. Hydrogen and chlorine combined is called hydrogen **chloride**
- For other combinations of non-metals as a general rule, the element that has a lower group number comes first in the name
 - E.g. carbon and oxygen combine to form CO₂ which is carbon dioxide since carbon is in Group 4 and oxygen in Group 6
- For compounds that contain certain groups of atoms:
 - There are **common groups** of atoms which occur regularly in chemistry
 - Examples include the carbonate ion (CO₃²⁻), sulfate ion (SO₄²⁻), hydroxide ion (OH⁻) and the nitrate ion (NO₃⁻)
 - When these ions form a compound with a metal atom, the name of the **metal** comes **first**
 - E.g. KOH is potassium hydroxide, CaCO₃ is calcium carbonate

Writing and balancing chemical equations

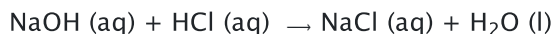
- Chemical equations use the chemical symbols of each reactant and product
- When balancing equations, there needs to be the **same number of atoms** of each element on either side of the equation
- The following non-metals must be written as diatomic molecules (i.e. molecules that contain two atoms): H₂, N₂, O₂, F₂, Cl₂, Br₂ and I₂
- Work across the equation from left to right, checking one element after another
- If there is a group of atoms, for example a nitrate group (NO₃⁻) that has not changed from one side to the other, then count the whole group as one entity



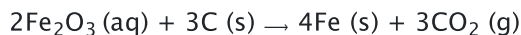
rather than counting the individual atoms.

◦ Examples of chemical equations:

▪ Acid-base neutralisation reaction:



▪ Redox reaction:



▪ In each equation there are equal numbers of each atom on either side of the reaction arrow so the equations are balanced

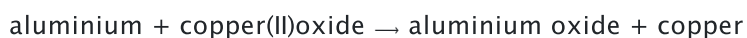
- The best approach is to practice lot of examples of balancing equations
- By trial and error change the coefficients (multipliers) in front of the formulae, one by one checking the result on the other side
- Balance elements that appear on their own, last in the process



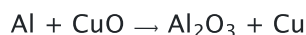
Worked Example

Example 1

Balance the following equation:



Unbalanced symbol equation:



STEP 1: ONLY 1 AL ATOM ON THE LEFT AND 2 ON THE RIGHT SO IF YOU END UP WITH 2, YOU MUST START WITH 2. CHANGE AL TO 2 AL



STEP 2: ONLY 1 O ATOM ON THE LEFT AND 3 ON THE RIGHT SO IF YOU END UP WITH 3, YOU MUST START WITH 3. TO ACHIEVE THIS, IT MUST BE 3CuO



STEP 3: THERE ARE 3 Cu ATOMS ON THE LEFT AND 1 ON THE RIGHT. THE ONLY WAY OF ACHIEVING 3 ON THE RIGHT IS TO HAVE 3 Cu

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Worked Example

Example 2:

Balance the following equation:



Unbalanced symbol equation:





STEP 1: THERE IS 1 Mg ATOM ON THE LEFT AND 1 ON THE RIGHT SO THERE ARE EQUAL NUMBERS ON BOTH SIDES SO THESE ARE KEPT THE SAME



STEP 2: THERE IS 1 O ATOM ON THE LEFT AND 1 ON THE RIGHT SO THERE IS AN EQUAL NUMBER ON BOTH SIDES. REMEMBER THAT YOU ARE COUNTING THE NITRATE GROUP AS A SEPARATE GROUP, SO DO NOT COUNT THE OXYGEN ATOMS IN THIS GROUP. BECAUSE THERE IS 1 NITRATE GROUP ON THE LEFT, BUT 2 NITRATE GROUPS ON THE RIGHT, THESE NEED TO BE BALANCED. SO, ADD A 2 IN FRONT OF THE HNO_3

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State symbols

- State symbols are written after each **formula** in chemical equations to show which physical state each substance is in
- Brackets are used and they are not usually subscripted although you may come across them written in this way
- Aqueous should remind you of the word 'aqua' and means the substance is **dissolved in water**
 - In other words it is a solution

SOLID	LIQUID	GAS	AQUEOUS
(s)	(l)	(g)	(aq)

The four state symbols show the physical state of substances at normal conditions

- Symbol equations should be included when writing chemical equations.
- An example of a reaction with state symbols is the reaction of copper carbonate with hydrochloric acid:



- Sometimes it can be hard to know what the correct state symbol is and we have to look for clues in the identity of substances in a reaction
- Generally, unless they are in a solution:
 - Metal compounds will always be solid, although there are a few exceptions
 - Ionic compounds will usually be solids
- Non-metal compounds could be solids, liquids or gases, so it depends on chemical structure
- Precipitates formed in solution count as solids
- In the worked examples above the final equations with the state symbols would be
 - $2\text{Al} (\text{s}) + 3\text{CuO} (\text{s}) \rightarrow \text{Al}_2\text{O}_3 (\text{s}) + 3\text{Cu} (\text{s})$
 - $\text{MgO} (\text{s}) + 2\text{HNO}_3 (\text{aq}) \rightarrow \text{Mg}(\text{NO}_3)_2 (\text{aq}) + \text{H}_2\text{O} (\text{l})$



Exam Tip

Be careful when writing the state symbol of solutions of liquids. For example, ethanol, or common alcohol, is a liquid at room temperature, so if it is pure alcohol then you would be using (l) as the state symbol; most of the time alcohol is used as a solution in water so (aq) is symbol to use.

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Deducing Symbol Equations

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EXTENDED

- For some reactions, you will not be given the unbalanced equation but you will be expected to use your knowledge learnt throughout the course to know or deduce the formula of compounds and then balance the equations



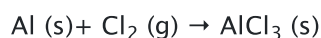
Worked Example

Aluminium burns in chlorine to form the white solid, aluminium chloride. Write the balanced symbol equation, including state symbols, for the reaction.

Answer:

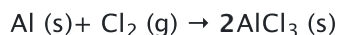
Step 1: Work out the formula and state symbols of the reactants and products to construct an unbalanced symbol equation:

- Aluminium is a solid metal, like other pure metals, it is an element so its formula is the same as its chemical symbol: **Al (s)**
- From your knowledge of Group VII elements, you should know that chlorine is a gas that exists as a diatomic molecule: **Cl₂ (g)**
- Aluminum chloride is a solid – this information is given in the question as you would not be expected to know this. Its formula is deduced from the charges on the ions present:
 - Aluminium has a 3+ charge and chloride ions have a 1- charge, therefore for the compound to be neutral, 3 chloride ions are needed for every 1 aluminium ion: **AlCl₃ (s)**
- The unbalanced symbol equation is thus:

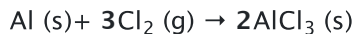


Step 2: Balance the equation:

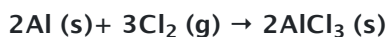
- Make the number of Cl on the RHS an even number by adding a 2 in front of AlCl₃:



- This gives 6 Cl on the RHS so now balance the number of Cl on the LHS by adding a 3 in front of Cl₂:

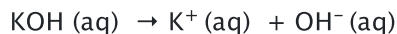
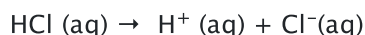


- Finally, there are now 2 Al on the RHS but only 1 on the LHS, so add a 2 in front of the Al on the LHS:



Balancing Ionic Equations

- In aqueous solutions ionic compounds **dissociate** into their ions, meaning they separate into the component ions that formed them
- E.g. hydrochloric acid and potassium hydroxide dissociate as follows:



- It is important that you can recognise common ionic compounds and their constituent ions
- These include:
 - Acids such as HCl and H₂SO₄
 - Group I and Group II hydroxides e.g. sodium hydroxide
 - Soluble salts e.g. potassium sulfate, sodium chloride
- Follow the example below to write ionic equations

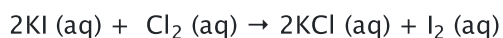


Worked Example

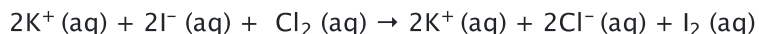
Write the ionic equation for the reaction of aqueous chlorine and aqueous potassium iodide.

Answer:

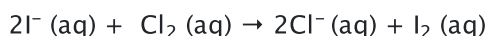
Step 1: Write out the full balanced equation:



Step 2: Identify the ionic substances and write down the ions separately



Step 3: Rewrite the equation eliminating the ions which appear on both sides of the equation (spectator ions) which in this case are the K⁺ ions:



Exam Tip

When balancing equations you **cannot** change any of the formulae, only the amount of each atom or molecule. This is done by changing the numbers that go in **front** of each chemical species.

YOUR NOTES



3.1.4 Ar & Mr

YOUR NOTES



Relative Masses

Relative Atomic Mass

- The symbol for the relative atomic mass is A_r
- The relative atomic mass for each element can be found in the Periodic Table along with the atomic number
- The relative atomic mass is shown underneath the atomic symbol and is larger than the atomic number (except for hydrogen where they are the same)
- Atoms are too small to accurately weigh but scientists needed a way to compare the masses of atoms
- The carbon-12 is used as the standard atom and has a fixed mass of 12 units
- It is against this atom which the masses of all other atoms are compared
- Relative atomic mass (A_r) can therefore be defined as:
 - The average mass of the isotopes of an element compared to 1/12th of the mass of an atom of ^{12}C**
- The relative atomic mass of carbon is 12
 - The relative atomic mass of magnesium is 24 which means that magnesium is twice as heavy as carbon
 - The relative atomic mass of hydrogen is 1 which means it has one twelfth the mass of one carbon-12 atom

Relative molecular (formula) mass

- The symbol for the relative molecular mass is M_r and it refers to the **total mass** of the molecule
- To calculate the M_r of a substance, you have to add up the relative atomic masses of all the atoms present in the formula
- Relative formula mass is used when referring to the total mass of an **ionic** compound

Relative Formula Mass Calculations Table

SUBSTANCE	ATOMS PRESENT	M_r
HYDROGEN (H_2)	$2 \times \text{H}$	$(2 \times 1) = 2$
WATER (H_2O)	$(2 \times \text{H}) + (1 \times \text{O})$	$(2 \times 1) + 16 = 18$
POTASSIUM CARBONATE (K_2CO_3)	$(2 \times \text{K}) + (1 \times \text{C}) + (3 \times \text{O})$	$(2 \times 39) + 12 + (3 \times 16) = 138$
CALCIUM HYDROXIDE ($\text{Ca}(\text{OH})_2$)	$(1 \times \text{Ca}) + (2 \times \text{O}) + (2 \times \text{H})$	$40 + (2 \times 16) + (2 \times 1) = 74$
AMMONIUM SULFATE ($(\text{NH}_4)_2\text{SO}_4$)	$(2 \times \text{N}) + (8 \times \text{H}) + (1 \times \text{S}) + (4 \times \text{O})$	$(2 \times 14) + (8 \times 1) + 32 + (4 \times 16) = 132$

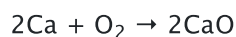
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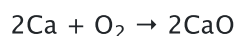
Reacting masses

- The Law of Conservation of mass tells us that mass cannot be created or destroyed
- In a chemical reaction, the total mass of reactants equals the total mass of the products
- We can use this, along with relative atomic/formula masses to perform calculations to identify the quantities of reactants or products involved in a chemical reaction

- **Example:**



- Relative atomic masses: Ca = 40; O = 16
- Using the balanced symbol equation shows that $2 \times 40 = 80$ units of mass of calcium react with $2 \times 16 = 32$ units of mass of oxygen to form $2 \times (40 + 16) = 112$ units of mass of CaO:



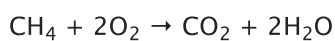
$$80 + 32 = 112$$

- The ratio of the mass of calcium and oxygen reacting will always be the same, regardless of the units
 - E.g. 80 g of calcium will react with 32 g of oxygen to form 112 g of calcium oxide
 - Or, 40 tonnes of calcium will react in excess oxygen to form 56 tonnes of calcium oxide



Worked Example

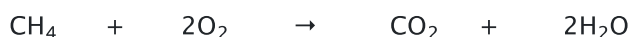
Calculate the mass of carbon dioxide produced when 32 g of methane, CH_4 , reacts completely in excess oxygen:



Relative atomic masses, A_r : H = 1; C = 12; O = 16

Answer

- In terms of relative mass the equation is:



$$12 + (4 \times 1) + 2 \times (2 \times 16) \rightarrow 12 + (2 \times 16) + 2 \times (2 \times 1 + 16)$$

$$16 + 64 \rightarrow 44 + 36$$

So 16 g of methane would react in excess oxygen to form 44 g of carbon dioxide

Therefore, 32 g of methane would produce $44 \times 2 = 88 \text{ g}$ of carbon dioxide

3.2 The Mole & the Avogadro Constant

3.2.1 The Mole

YOUR NOTES



The Mole & the Avogadro Constant

EXTENDED

The Mole & Avogadro's Constant

- Chemical amounts are measured in moles
- The mole, symbol mol, is the SI unit of **amount of substance**
- One mole of a substance contains the same number of the stated particles, atoms, molecules, or ions as one mole of any other substance
- One mole contains **6.02×10^{23}** particles (e.g. atoms, ions, molecules); this number is known as the **Avogadro Constant**
- For example:
 - One mole of sodium (Na) contains 6.02×10^{23} atoms of sodium
 - One mole of hydrogen (H₂) contains 6.02×10^{23} molecules of hydrogen
 - One mole of sodium chloride (NaCl) contains 6.02×10^{23} formula units of sodium chloride
- The mass of 1 mole of a substance is known as the **molar mass**
- For an element, it is the same as the **relative atomic mass** written in grams
- For a compound it is the same as the **relative formula mass** or **relative molecular mass** in grams

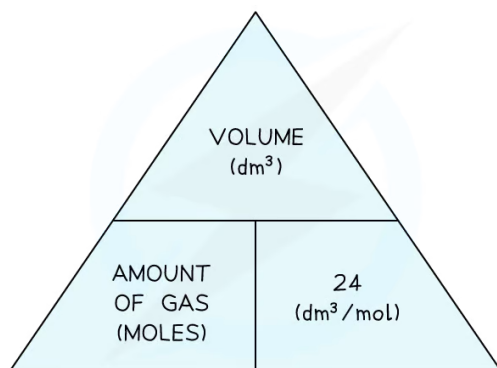
The Mole & Volume of Gas

YOUR NOTES



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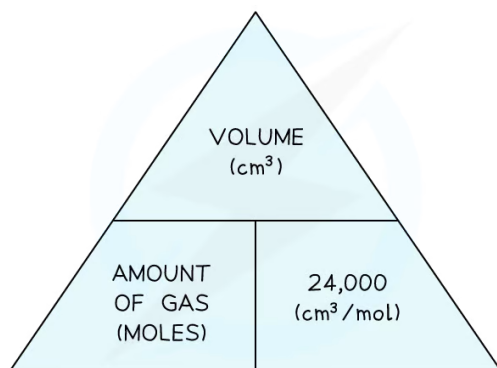
- **Avogadro's Law** states that at the same conditions of **temperature** and **pressure**, equal amounts of gases occupy the **same volume** of space
- At room temperature and pressure, the volume occupied by one mole of any gas was found to be **24 dm³** or **24,000 cm³**
- This is known as the **molar gas volume at RTP**
- RTP stands for "room temperature and pressure" and the conditions are **20 °C** and **1 atmosphere** (atm)
- From the molar gas volume the following formula triangle can be derived:



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Formula triangle showing the relationship between moles of gas, volume in dm³ and the molar volume

- If the volume is given in cm³ instead of dm³, then divide by 24,000 instead of 24:



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Formula triangle showing the relationship between moles of gas, volume in cm³ and the molar volume

- The formula can be used to calculate the number of moles of gases from a given volume or vice versa
- Simply cover the one you want and the triangle tells you what to do

To find the volume of a gas

$$\text{Volume} = \text{Moles} \times \text{Molar Volume}$$

Examples of Converting Moles into Volumes Table

NAME OF GAS	AMOUNT OF GAS	VOLUME OF GAS
HYDROGEN	3 mol	$(3 \times 24) = 72 \text{ dm}^3$
CARBON DIOXIDE	0.25 mol	$(0.25 \times 24) = 6 \text{ dm}^3$
OXYGEN	5.4 mol	$(5.4 \times 24,000) = 129,600 \text{ cm}^3$
AMMONIA	0.02 mol	$(0.02 \times 24) = 0.48 \text{ dm}^3$

To find the moles of a gas

$$\text{Moles} = \text{Volume} \div \text{Molar Volume}$$

Examples of Converting Volumes into Moles Table

NAME OF GAS	VOLUME OF GAS	MOLES OF GAS
METHANE	225.6 dm^3	$(225.6 : 24) = 9.4 \text{ mol}$
CARBON MONOXIDE	7.2 dm^3	$(7.2 : 24) = 0.3 \text{ mol}$
SULFUR DIOXIDE	960 dm^3	$(960 : 24) = 40 \text{ mol}$
OXYGEN	1200 cm^3	$(1200 : 24000) = 0.05 \text{ mol}$

YOUR NOTES



3.2.2 Linking Moles, Mass & Mr

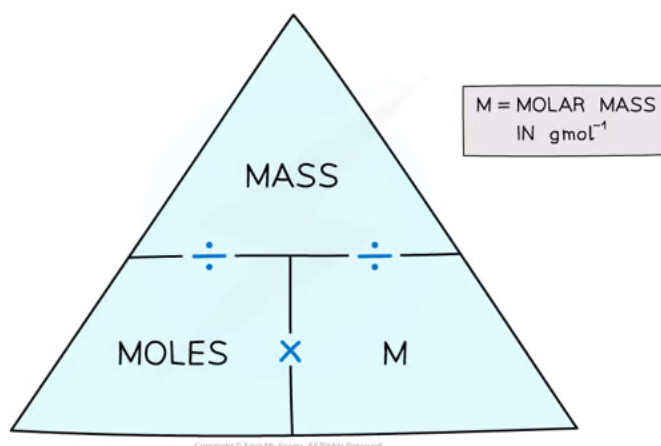
YOUR NOTES



Linking Moles, Mass & Mr

EXTENDED

- Although elements and chemicals react with each other in **molar ratios**, in the laboratory we use digital balances and grams to measure quantities of chemicals as it is impractical to try and measure out moles
- Therefore we have to be able to convert between moles and grams
- We can use the following formula to convert between moles, mass in grams and the molar mass
- The mass of 1 mole of a substance is known as the **molar mass**
- For an element, it is the same as the **relative atomic mass** written in grams
- For a compound it is the same as the **relative formula mass** or **relative molecular mass** in grams



Formula triangle for moles, mass and molar mass



Worked Example

What is the mass of 0.250 moles of zinc?

Answer:

- From the Periodic Table, the relative atomic mass of Zn is 65.4
- So, the molar mass is 65.4 g/mol
- The mass is calculated by **moles x molar mass**
- This comes to 0.250 mol x 65.4 g/mol = **16.3 g**



Worked Example

How many moles are in 2.64 g of sucrose, $C_{12}H_{22}O_{11}$ ($M_r = 342.3$)?

Answer:

- The molar mass of sucrose is 342.3 g/mol
- The number of moles is found by **mass ÷ molar mass**
- This comes to $2.64 \text{ g} \div 342.3 \text{ g/mol} = 7.71 \times 10^{-3} \text{ mol}$



Worked Example

In 15.7 g of water ($M_r = 18$):

- a. How many molecules are there?
- b. How many atoms are there?

Answer – part a

- The molar mass of water is 18 g/mol
- The number of moles is found by **mass ÷ molar mass**
- This comes to $15.7 \text{ g} \div 18 \text{ g/mol} = 0.872 \text{ mol}$
- There are 6.02×10^{23} molecules of water in 1 mole of water
- In 0.872 moles of water, there are $6.02 \times 10^{23} \times 0.872 = 5.25 \times 10^{23}$ molecules

Answer – part b

- In each molecule of water there are 3 atoms (2 hydrogen atoms, one oxygen atom)
- The number of atoms in 15.7 g = $3 \times 5.25 \times 10^{23} = 1.58 \times 10^{24}$ atoms



Exam Tip

Always show your workings in calculations as its easier to check for errors and you may pick up credit if you get the final answer wrong.

YOUR NOTES



3.2.3 Reacting Masses

YOUR NOTES



Reacting Masses

EXTENDED

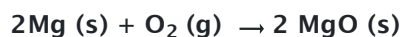
- Chemical equations can be used to calculate the **moles** or **masses** of reactants and products
- To do this, information given in the question is used to find the amount in moles of the substances being considered
- Then, the **ratio** between the substances is identified using the balanced chemical equation
- Once the moles have been determined they can then be converted into grams using the relative atomic or relative formula masses



Worked Example

Example 1

Calculate the mass of magnesium oxide that can be made by completely burning 6.0 g of magnesium in oxygen in the following reaction:



Relative formula masses (M_r): $\text{Mg} = 24$; $\text{MgO} = 40$

$$\text{MOLES} = \frac{\text{MASS}}{\text{MOLAR MASS}}$$

$$\text{MOLES} = \frac{6}{24} = 0.25$$



RATIO IS THUS 1:1

SO 0.25 MOLES Mg
PRODUCES 0.25 MOLES MgO

$$\text{MASS} = \text{MOLES} \times \text{MOLAR MASS}$$

$$\text{MASS} = 0.25 \times 40 = 10\text{g}$$

STEP 1: CALCULATE THE MOLES
OF MAGNESIUM USED IN THE
REACTION

STEP 2: FIND THE RATIO OF
Mg TO MgO USING THE MOLAR
RATIO FROM THE BALANCED
EQUATION

STEP 3: CALCULATE THE MASS
OF MgO



Worked Example

Example 2

Calculate the mass of aluminium, in tonnes, that can be produced from 51 tonnes of aluminium oxide. The equation for the reaction is:



Relative formula masses (M_r): $\text{Al} = 27$; $\text{Al}_2\text{O}_3 = 102$



$$\text{MASS OF AL}_2\text{O}_3 \text{ IN g} = 51 \times 10^6 \\ = 51,000,000 \text{ g}$$

STEP 1: CALCULATE THE MOLES OF ALUMINIUM OXIDE USED

$$\text{MOLES} = \frac{\text{MASS}}{\text{MOLAR MASS}}$$

$$\text{MOLES} = \frac{51,000,000}{102} = 500,000$$



RATIO IS THUS 1 : 2

SO 500,000 MOLES Al_2O_3 PRODUCES 1,000,000 MOLES AL

STEP 2: FIND THE RATIO OF Al_2O_3 TO AL USING THE MOLAR RATIO FROM BALANCED EQUATION

$$\text{MASS} = \text{MOLES} \times \text{MOLAR MASS}$$

$$\text{MASS} = 1,000,000 \times 27 \\ = 27,000,000 \text{ g}$$

STEP 3: CALCULATE THE MASS OF AL



Exam Tip

Remember molar ratio of a balanced equation gives you the ratio of the amounts of each substance in the reaction.

Limiting Reactants

- A chemical reaction stops when one of the reactants is used up
- The reactant that is used up first is the **limiting reactant**, as it limits the duration and hence the amount of product that a reaction can produce
- The amount of product is therefore **directly proportional** to the amount of the limiting reactant added at the beginning of a reaction
- The limiting reactant is the reactant which is **not present in excess** in a reaction
- In order to determine which reactant is the limiting reactant in a reaction, we have to consider the ratios of each reactant in the balanced equation
- When performing reacting mass calculations, the limiting reactant is always the number that should be used as it indicates the maximum possible amount of product
- The steps are:
 1. Write the balanced equation for the reaction
 2. Calculate the moles of each reactant
 3. Compare the moles & deduce the limiting reactant



Worked Example

9.2 g of sodium is reacted with 8.0 g of sulfur to produce sodium sulfide, Na_2S .

Which reactant is in excess and which is the limiting reactant?

Relative atomic masses (A_r): Na = 23; S = 32

Answer:

Step 1: Write the balanced equation and determine the molar ratio

$2\text{Na} + \text{S} \rightarrow \text{Na}_2\text{S}$ so the molar ratio of Na : S is 2 : 1

Step 2: Calculate the moles of each reactant

$$\text{Moles} = \text{Mass} \div \text{Molar Mass}$$

$$\text{Moles Na} = 9.2 \div 23 = 0.40$$

$$\text{Moles S} = 8.0 \div 32 = 0.25$$

Step 3: Compare the moles

- To react completely 0.40 moles of Na requires 0.20 moles of S and since there are 0.25 moles of S, then S is in excess
- **Na is therefore the limiting reactant**

YOUR NOTES



3.2.4 Calculating Concentration

Units of Concentration

- A solid substance that dissolves in a liquid is called a solute, the liquid is called a solvent and the two when mixed together form a solution
- Most chemical reactions occur between solutes which are dissolved in solvents, such as water or an organic solvent
- Concentration simply refers to the amount of solute there is in a specific volume of the solvent
- The greater the amount of solute in a given volume, the greater the concentration
- The amount of solute can be expressed in grams or moles
- Typically, concentration is expressed in terms of the amount of substance per dm^3 , therefore the units of concentration are either: **g/dm^3** or **mol/dm^3**

YOUR NOTES



Calculating Concentration

YOUR NOTES



EXTENDED

- **Concentration** refers to the amount of solute there is in a specific volume of the solvent
- A general formula to calculate the concentration in g/dm^3 is:

$$\text{concentration (g/dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{volume of solution (dm}^3\text{)}}$$

- Concentration can be measured in grams per cubic decimetre
- 1 decimetre cubed (dm^3) = 1000 cm^3
 - 1 decimetre cubed (dm^3) is the same as 1 litre
- You may be given data in a question which needs to be converted from cm^3 to dm^3 or the other way around
 - To go from cm^3 to dm^3 :
 - Divide by 1000
 - To go from dm^3 to cm^3 :
 - Multiply by 1000



Worked Example

A student dissolved 10 g of sodium hydroxide, NaOH, in 2 dm^3 of distilled water. Calculate the concentration of the solution in g/dm^3 .

Answer:

MASS: 10g

VOLUME OF SOLUTION: 2 dm^3

STEP 1: WRITE DOWN
THE INFORMATION
GIVEN IN THE QUESTION

$$\text{CONCENTRATION} = \frac{\text{MASS OF SOLUTE IN g}}{\text{VOLUME IN dm}^3}$$

$$\text{CONCENTRATION} = \frac{10\text{g}}{2\text{dm}^3} = 5 \text{ g/dm}^3$$

STEP 2: SUBSTITUTE THE
VALUES INTO THE
EQUATION AND SOLVE

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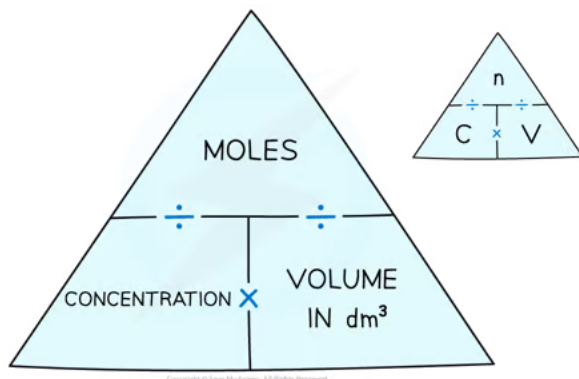
Calculating Concentration using Moles

- It is more useful to a chemist to express concentration in terms of moles per unit volume rather than mass per unit volume
- Concentration can therefore be expressed in moles per decimetre cubed and calculated using the following equation:

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles of solute (mol)}}{\text{volume of solution (dm}^3\text{)}}$$

- We can modify the concentration formula to include moles
 - The units in the answer can be written as mol/dm^3 (this can also be written as mol dm^{-3})
- You may have to convert from g/dm^3 into mol/dm^3 and vice versa depending on the question

- To go from g/dm^3 to mol/dm^3
 - Divide by the molar mass in grams
- To go from mol/dm^3 to g/dm^3
 - Multiply by the molar mass in grams
- Some students find formula triangles help them to understand the relationship:



The concentration-moles formula triangle can help you solve these problems

? Worked Example

Example 1

Calculate the amount of solute, in moles, present in 2.5 dm^3 of a solution whose concentration is 0.2 mol/dm^3 .

Answer 1

CONCENTRATION
OF SOLUTION: 0.2 mol/dm^3

VOLUME OF SOLUTION: 2.5 dm^3

MOLES OF SOLUTE = 0.2×2.5
= 0.5 mol

STEP 1: WRITE DOWN
THE INFORMATION
GIVEN IN THE QUESTION

STEP 2: SUBSTITUTE THE
VALUES INTO THE
EQUATION AND SOLVE

? Worked Example

Example 2

Calculate the concentration of a solution of sodium hydroxide, NaOH , in mol/dm^3 , when 80 g is dissolved in 500 cm^3 of water.

Relative atomic masses, A_r : $\text{Na} = 23$; $\text{H} = 1$; $\text{O} = 16$

Answer 2

YOUR NOTES



$$\begin{aligned} \text{Mr NaOH} &= 23 + 16 + 1 \\ &= 40 \text{ g} \end{aligned}$$

$$\begin{aligned} \text{SO } 40 \text{ g} &= 1 \text{ MOLE,} \\ \text{THUS } 80 \text{ g} &= 2 \text{ MOLES} \end{aligned}$$

$$\frac{500}{1000} = 0.5 \text{ dm}^3$$

$$\begin{aligned} \text{CONCENTRATION} &= \frac{2 \text{ mol}}{0.5 \text{ dm}^3} \\ &= 4 \text{ mol/dm}^3 \end{aligned}$$

STEP 1: CONVERT THE GRAMS OF NaOH TO MOLES

STEP 2: CONVERT cm^3 TO dm^3 FOR H_2O

STEP 3: SUBSTITUTE THE VALUES INTO THE EQUATION AND SOLVE FOR CONCENTRATION

YOUR NOTES



Worked Example

Example 3

25.0 cm^3 of 0.050 mol/dm^3 sodium carbonate was completely neutralised by 20.00 cm^3 of dilute hydrochloric acid. Calculate the concentration in mol/dm^3 of the hydrochloric acid.

Answer 3

$$\begin{aligned} \text{AMOUNT OF} \\ \text{Na}_2\text{CO}_3 &= (25.0 \times 0.050) \div 1000 \\ &= 0.00125 \text{ mol} \end{aligned}$$

STEP 1: CALCULATE THE AMOUNT, IN MOLES, OF SODIUM CARBONATE REACTED BY REARRANGING THE EQUATION FOR AMOUNT OF SUBSTANCE (mol) AND DIVIDING BY 1000 TO CONVERT cm^3 TO dm^3

1 mol OF Na_2CO_3 REACTS WITH 2 mol OF HCl, SO THE MOLAR RATIO IS 1:2

STEP 2: CALCULATE THE AMOUNT, IN MOLES, OF HYDROCHLORIC ACID REACTED

THEREFORE 0.00125 moles OF Na_2CO_3 REACT WITH 0.00250 moles OF HCl

$$\begin{aligned} 1 \text{ dm}^3 &= 1000 \text{ cm}^3 \\ \text{VOLUME OF HCl} &= 20 \div 1000 \\ &= 0.0200 \text{ dm}^3 \end{aligned}$$

STEP 3: CALCULATE THE CONCENTRATION, IN mol/dm^3 OF THE HYDROCHLORIC ACID

$$\begin{aligned} \text{CONCENTRATION HCl (mol/dm}^3) &= \\ 0.00250 \div 0.0200 &= 0.125 \text{ mol/dm}^3 \end{aligned}$$



Exam Tip

Remember to always convert the units from cm^3 to dm^3 by dividing by 1000.

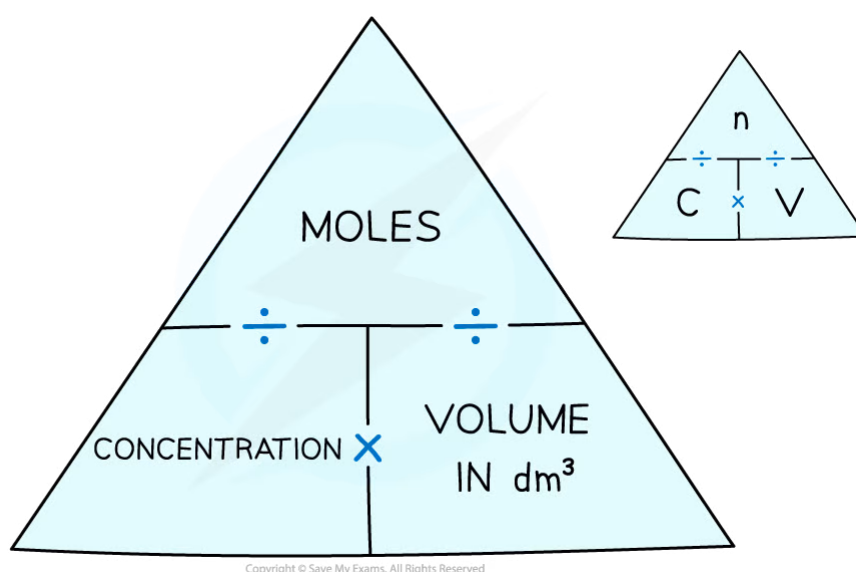
3.2.5 Titration Calculations

YOUR NOTES



Titration Calculations

- Titrations are a method of analysing the **concentration** of solutions
- Acid–base titrations are one of the most important kinds of titrations
- They can determine exactly how much alkali is needed to neutralise a quantity of acid – and vice versa
- You may be asked to calculate the **moles** present in a given amount, the **concentration** or **volume** required to **neutralise** an acid or a base
- Once a titration is completed and the average titre has been calculated, you can now proceed to calculate the unknown variable using the formula triangle as shown below



Formula triangle showing the relationship between concentration, number of moles and volume of liquid

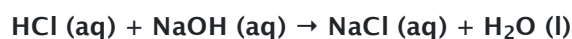


Worked Example

A solution of 25.0 cm³ of hydrochloric acid was titrated against a solution of 0.100 mol/dm³ NaOH and 12.1 cm³ were required for complete reaction. Determine the concentration of the acid.

Answer:

Step 1: Write the equation for the reaction:



Step 2: Calculate the number of moles of the NaOH

$$\text{Moles} = (\text{volume} \div 1000) \times \text{concentration}$$

$$\text{Moles of NaOH} = 0.012 \text{ dm}^3 \times 0.100 \text{ mol/dm}^3 = 1.21 \times 10^{-3} \text{ mol}$$

Step 3: Deduce the number of moles of the acid

Since the acid reacts in a 1:1 ratio with the alkali, the number of moles of HCl is also $1.21 \times 10^{-3} \text{ mol}$

This is present in 25.0 cm^3 of the solution ($25.0 \text{ cm}^3 = 0.025 \text{ dm}^3$)

Step 4: Find the concentration of the acid

Concentration = moles $\div$ volume (dm^3)

$$\text{Concentration of HCl} = 1.21 \times 10^{-3} \text{ mol} \div 0.025 \text{ dm}^3 = \mathbf{0.0484 \text{ mol/dm}^3}$$

YOUR NOTES



3.2.6 Empirical & Molecular Formula

YOUR NOTES



Calculating Empirical & Molecular Formulae

EXTENDED

Calculating Empirical Formula

- The **empirical formula** is the simplest whole number ratio of the atoms of each element present in one molecule or formula unit of the compound
 - E.g. the empirical formula of ethanoic acid is CH_2O
- Organic molecules** often have **different** empirical and molecular formulae
- The formula of an **ionic compound** is always an **empirical formula**



Worked Example

A compound that contains 10 g of hydrogen and 80 g of oxygen.

What is its empirical formula of this compound?

Answer:

	hydrogen	oxygen
Write the mass of each element	10 g	80 g
Divide each mass by the relative atomic mass to find the number of moles	$10/1 = 10$	$80/16 = 5$
Find the molar ratio by dividing by the smallest number	$10/5 = 2$	$5/5 = 1$

Empirical formula = H_2O



Worked Example

Substance X was analysed and found to contain 31.58% carbon, 5.26% hydrogen and 63.16% oxygen by mass.

What is the empirical formula of substance X?

Relative atomic masses, A_r : C = 12; H = 1; O = 16

Answer:

	carbon	hydrogen	oxygen
Convert % to g by assuming 100 g of substance is present	31.58 g	5.26 g	63.16 g
Divide each mass by the relative atomic mass to find the number of moles in 100 g	$31.58/12 = 2.63$	$5.26/1 = 5.26$	$63.16/16 = 3.95$
Find the molar ratio by dividing by the smallest number	$2.63/2.63 = 1$	$5.26/2.63 = 2$	$3.95/2.63 = 1.5$



Multiply all by 2 to obtain a whole number ratio	2	4	3
--	---	---	---

Empirical formula = $C_2H_4O_3$



Exam Tip

The molar ratio must be a whole number. If you don't get a whole number when calculating the ratio of atoms in an empirical formula, such as 1.5, multiply that and the other ratios to achieve whole numbers.

Calculating Molecular Formula

- Molecular formula gives the **actual numbers of atoms of each element** present in the formula of the compound
- To calculate the molecular formula:
 - Step 1: Find the relative formula mass of the empirical formula
 - Step 2: Use the following equation:

$$\frac{\text{relative formula mass of molecular formula}}{\text{relative formula mass of empirical formula}}$$

- Step 3: Multiply the number of each element present in the empirical formula by the number from step 2 to find the molecular formula

Table showing the Relationship between Empirical and Molecular Formula

NAME OF COMPOUND	EMPIRICAL FORMULA	MOLECULAR FORMULA
METHANE	CH_4	CH_4
ETHANE	CH_3	C_2H_6
ETHENE	CH_2	C_2H_4
BENZENE	CH	C_6H_6



Worked Example

The empirical formula of X is $C_4H_{10}S_1$

The relative formula mass (M_r) of X is 180.

What is the molecular formula of X?

(Relative atomic mass, A_r : Carbon : 12 Hydrogen : 1 Sulfur : 32)

Answer



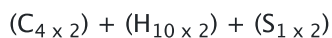
Step 1 – Calculate the relative formula mass of the empirical formula

$$M_r = (12 \times 4) + (1 \times 10) + (32 \times 1) = 90$$

Step 2 – Divide relative formula mass of X by relative formula mass of empirical formula

$$180 / 90 = 2$$

Step 3 – Multiply each number of elements by 2



Molecular Formula of X = $C_8H_{20}S_2$

Deducing formulae of hydrated salts

- The formula of hydrated salts can be determined experimentally by weighing a sample of the hydrated salt, heating it until the water of crystallisation has been driven off, then reweighing the now anhydrous salt
- From the results, you can determine the mass of anhydrous salt and the mass of the water of crystallisation
- Applying a similar approach to deducing empirical formulae, the formula of the hydrated salt can be calculated



Worked Example

11.25 g of hydrated copper sulfate, $CuSO_4 \cdot xH_2O$, is heated until it loses all of its water of crystallisation. It is reweighed and its mass is 7.19 g. What is the formula of the hydrated copper(II) sulfate?

	$CuSO_4$	H_2O
Deduce the mass of water of crystallisation and anhydrous salt	= mass of salt after heating = 7.19 g	mass of hydrated salt – mass of anhydrous salt: 11.25 – 7.19 = 4.06 g
Divide each mass by the relative formula mass to find the number of moles	$\frac{7.19}{160} = 0.045$	$\frac{4.06}{18} = 0.226$
Find the molar ratio by dividing by the smallest number	$\frac{0.045}{0.045} = 1$	$\frac{0.226}{0.045} = 5$

Therefore the formula of hydrated copper(II) sulfate is $CuSO_4 \cdot 5H_2O$



Exam Tip

The specification is not clear about whether deducing the formula of hydrated salts is required, however, it is an application of deducing empirical formulae so it is worth knowing how to do this.

YOUR NOTES





3.2.7 Percentage Yield & Purity

Calculating Percentage Yield, Percentage by Mass & Percentage Purity

EXTENDED

Percentage Yield

- **Yield** is the term used to describe the amount of **product** you get from a reaction
- In practice, you **never** get 100% yield in a chemical process for several reasons
- These include:
 - Some reactants may be left behind in the **equipment**
 - The reaction may be **reversible** and in these reactions a high yield is never possible as the products are continually turning back into the reactants
 - Some products may also be lost during **separation** and **purification** stages such as filtration or distillation
 - There may be **side reactions** occurring where a substance reacts with a gas in the air or an **impurity** in one of the reactants
 - Products can also be lost during **transfer** from one container to another

Actual & Theoretical Yield

- The **actual yield** is the recorded amount of product obtained
- The **theoretical yield** is the amount of product that would be obtained under perfect practical and chemical conditions
- It is calculated from the balanced equation and the reacting masses
- The **percentage yield** compares the actual yield to the theoretical yield
- For economic reasons, the objective of every chemical producing company is to have as high a percentage yield as possible to increase profits and reduce costs and waste
- To calculate percentage yield the following equation is used:

$$\text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$



Worked Example

Copper(II) sulfate may be prepared by the reaction of dilute sulfuric acid on copper(II) oxide.

A student prepared 1.6 g of dry copper(II) sulfate crystals.

Calculate the percentage yield if the theoretical yield is 2.0 g.

Answer

- Actual yield of copper(II) sulfate = 1.6 g
- Percentage yield of copper(II) sulfate = $(1.6 / 2.0) \times 100$
- Percentage yield = **80%**



Exam Tip

Percentage yield cannot be greater than 100%, if you calculate a percentage higher than this, you have made an error! The most common error is to divide the theoretical yield by the actual yield so you may just need to swap the number around in your calculation.

YOUR NOTES



Calculating percentage mass

- You may be asked to find the percentage by mass of an element within a compound
- This can be found by calculating the relative formula mass of the compound and then using the following equation:

$$\% \text{ of an element} = \frac{\text{total mass of the element in the compound}}{\text{relative formula mass of the compound}} \times 100$$



Worked Example

Calculate the percentage by mass of iron in iron(III) oxide, Fe_2O_3 .

Relative atomic masses, A_r : Fe = 56 O = 16

Answer:

Step 1: Calculate the relative formula mass of the compound

$$(2 \times 56) + (3 \times 16) = 160$$

Step 2: Work out the total A_r of the element you have been asked about in the question, in this case, iron

$$2 \times 56 = 112$$

Step 3: Divide the total A_r of the element by the M_r of the compound, then multiply by 100.

$$(112 \div 160) \times 100 = 70\%$$

Percentage Purity

- A pure substance has nothing else mixed with it
- Often, the product you are trying to obtain may become contaminated with unwanted substances such as unreacted reactants, catalysts and other impurities
- To calculate the percentage purity the following equation is used:

$$\% \text{ purity} = \frac{\text{mass of pure substance}}{\text{total mass of substance}} \times 100$$

YOUR NOTES



Worked Example

A sample of lead(II) bromide was made. It weighed 15 g.

The sample was found to be impure and only contained 13.5 g of lead(II) bromide.

Calculate the percentage purity of the lead(II) bromide.

Answer:

- The total mass of the substance is 15 g
- The mass of the pure substance is 13.5 g
- To calculate the percentage purity: $(13.5 / 15) \times 100$
- Percentage purity = **90 %**



Exam Tip

All of these calculations are to find a percentage so don't forget to multiply by 100 to convert your answer to a percentage.

